

Determination of refractive index — Solution

W23 (2023) — English translation

A1

0.41 pts

For each x , find n .

Answer: See A6.

A2

0.82 pts

For each x , find λ .

Answer: See A6.

A3

0.82 pts

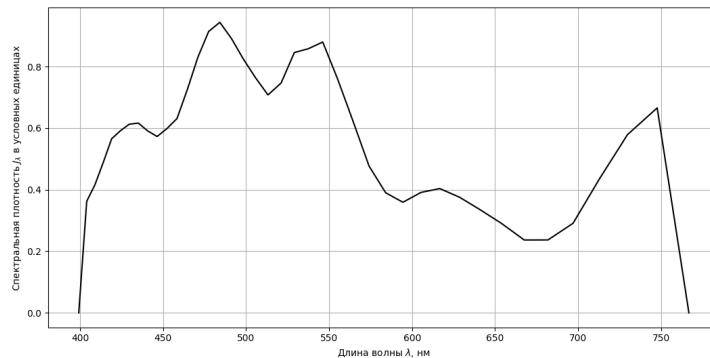
For each x , find $\mathcal{J}_\lambda$ (in arbitrary units per nm). Since for each x we have already calculated λ , we obtain the dependence $\mathcal{J}_\lambda(\lambda)$.

Answer: See A6.

A4

0.49 pts

Plot a graph of $\mathcal{J}_\lambda(\lambda)$.



A6

1.23 pts

Find for each x the value σ . Since for each x we found a λ , we also obtained the dependence $\lambda(\sigma)$.

For each x_i we find the value of the sum

$$\Sigma_i = 2 \sum_{k=1}^{i-1} \mathcal{J}_{x_k} + \mathcal{J}_{x_i},$$

as well as the normalizing factor

$$\Sigma = 2 \sum_i \mathcal{J}_{x_i}.$$

Since $\lambda(x)$ is a decreasing function, then $\sigma = 0$ corresponds to the value $x = x_{\max}$, therefore σ_i is calculated by the formula:

$$\sigma_i = 1 - \frac{\Sigma_i}{\Sigma}.$$

Answer: $x \quad \mathcal{J}_x \quad n \quad \lambda, \text{ nm} \quad \mathcal{J}_\lambda \quad \sigma$

0.1250	0	1.2862	766.8	0.000	1.0000	0.1275	4917	1.2938	747.7	0.6658	0.9684
0.1300	4024	1.3014	729.8	0.5789	0.9110	0.1325	2854	1.3091	712.9	0.4351	0.8669
0.1350	1805	1.3170	697.0	0.2909	0.8370	0.1375	1393	1.3249	681.9	0.2368	0.8165
0.1400	1322	1.3329	667.6	0.2366	0.7990	0.1425	1545	1.3411	653.9	0.2905	0.7807
0.1450	1701	1.3493	640.9	0.3354	0.7598	0.1475	1819	1.3576	628.6	0.3756	0.7372
0.1500	1870	1.3661	616.7	0.4037	0.2135	0.1525	1736	1.3746	605.4	0.3912	0.6904
0.1550	1529	1.3833	594.5	0.3592	0.6694	0.1575	1595	1.3920	584.1	0.3901	0.6494
0.1600	1875	1.4009	574.1	0.4768	0.6271	0.1625	2360	1.4099	564.4	0.6233	0.5999
0.1650	2775	1.4190	555.1	0.7604	0.5669	0.1675	3098	1.4283	546.2	0.8798	0.5292
0.1700	2918	1.4376	537.5	0.8580	0.4906	0.1725	2780	1.4471	529.1	0.8456	0.4511
0.1750	2375	1.4567	521.1	0.7465	0.4210	0.1775	2181	1.4664	513.2	0.7079	0.3917
0.1800	2281	1.4763	505.7	0.7638	0.3631	0.1825	2390	1.4863	498.3	0.8251	0.3331
0.1850	2506	1.4964	491.2	0.8912	0.3017	0.1875	2578	1.5067	484.2	0.9438	0.2690
0.1900	2428	1.5171	477.5	0.9144	0.2369	0.1925	2146	1.5277	471.0	0.8308	0.2075
0.1950	1823	1.5384	464.6	0.7251	0.1821	0.1975	1546	1.5493	458.4	0.6313	0.1604
0.2000	1428	1.5603	452.3	0.5984	0.1414	0.2025	1333	1.5714	446.4	0.5729	0.1236
0.2050	1342	1.5828	440.7	0.5911	0.1065	0.2075	1366	1.5942	435.1	0.6164	0.0891
0.2100	1326	1.6059	429.6	0.6127	0.0718	0.2125	1251	1.6177	424.3	0.5916	0.0553
0.2150	1169	1.6297	419.0	0.5654	0.0397	0.2175	986	1.6419	413.9	0.4876	0.0259
0.2200	822	1.6542	408.9	0.4155	0.0143	0.2225	702	1.6668	404.0	0.3625	0.0045
0.2250	0	1.6795	399.2	0.0000	0.0000						

B1

0.41 pts

For each x , find n .

Answer: See **B3**.

B2

1.23 pts

Find for each x the value σ .

We act similarly to **A6**.

Answer: See **B3**.

B3

4.10 pts

Using linear interpolation, find which values of λ correspond to the resulting values of σ . Since n were calculated in **B1**, we obtain the desired dependence $n(\lambda)$.

Answer: $x' \quad \mathcal{J}_{x'} \quad n \quad \sigma \quad \lambda$

0.050	0	1.0961	1.0000	766.8	0.055	3746	1.1069	0.9752	751.8	0.060	3194	1.1179
0.9292	735.5	0.065	2647	1.1292	0.8905	722.0	0.070	2040	1.1407	0.8595	709.0	0.075
1419	1.1525	0.8365	696.7	0.080	1073	1.1645	0.8200	684.5	0.085	1111	1.1768	0.8056
672.9	0.090	1240	1.1894	0.7900	660.9	0.095	1365	1.2023	0.7727	649.0	0.100	1499
1.2155	0.7538	637.6	0.105	1609	1.2290	0.7332	626.5	0.110	1622	1.2428	0.7118	615.9
0.115	1572	1.2569	0.6906	605.5	0.120	1370	1.2714	0.6711	595.4	0.125	1450	1.2862
0.6524	585.7	0.130	1543	1.3014	0.6326	576.5	0.135	2023	1.3170	0.6090	567.6	0.140
2506	1.3329	0.5790	558.5	0.145	2893	1.3493	0.5432	549.5	0.150	2974	1.3661	0.5043
540.6	0.155	2620	1.3833	0.4673	532.2	0.160	2532	1.4009	0.4332	524.0	0.165	2168
1.4190	0.4020	516.0	0.170	2073	1.4376	0.3739	508.5	0.175	2274	1.4567	0.3451	501.2
0.180	2652	1.4763	0.3125	493.6	0.185	2744	1.4964	0.2767	485.9	0.190	2514	1.5171
0.2419	478.6	0.195	2427	1.5384	0.2092	471.3	0.200	1927	1.5603	0.1803	464.1	0.205
1570	1.5828	0.1572	457.3	0.210	1496	1.6059	0.1369	450.8	0.215	1551	1.6297	0.1167
444.1	0.220	1620	1.6542	0.0957	437.2	0.225	1589	1.6795	0.0744	430.4	0.230	1455
1.7055	0.0542	423.9	0.235	1339	1.7323	0.0357	417.6	0.240	1107	1.7600	0.0195	411.2
0.245	921	1.7885	0.0061	404.8	0.250	0	1.8179	0.0000	399.2			

B4

0.49 pts

Plot $n(\lambda)$ at a convenient scale (the curve should take up most of the graph on both axes; the axes do not have to start at zero).

